

GALLANT IAS ACADEMY
PRELIMS CUM MAINS BATCH 2027
CSAT

DIGITS & PLACE VALUE, MATHEMATICAL OPERATION

1. The number of times the digit 5 will appear while writing the integers from 1 to 1000 is:
a) 269
b) 271
c) 300
d) 302
2. How many times does 5 appear in all two-digit positive integers ?
a) 18
b) 19
c) 20
d) 21
3. For any choices of values of X, Y and Z, the 6-digit number of the form XYZXYZ is divisible by :
(a) 7 and 11 only
(b) 11 and 13 only
(c) 7 and 13 only
(d) 7, 11 and 13
4. Let XYZ be a three-digit number, where $(X+Y+Z)$ is not a multiple of 3. Then $(XYZ+YZX+ZXY)$ is not divisible by
a) 3
b) 9
c) 37
d) $(X+Y+Z)$
5. There are certain 2-digit numbers. The difference between the number and the one obtained on reversing it is always 27. How many such maximum 2-digit numbers are there?
a) 3
b) 4
c) 5
d) None of the above
6. A is a 2-digit number with different digits. B is also a 2-digit number and is obtained by reversing the digits of A. If $A-B$ is a multiple of 27, where $A > B$, how many such different A's are possible?
a) 6
b) 9
c) 12
d) 18
7. Let X be a two-digit number and Y be another two-digit number formed by interchanging the digits of X. If $(X+Y)$ is the greatest two-digit number, then what is the number of possible values of X?

- a) 2
b) 4
c) 6
d) 8
8. The ratio of a two-digit natural number to a number formed by reversing its digits is 4:7. The number of such pairs is
a) 5
b) 4
c) 3
d) 2
9. A 3-digit number ABC, on multiplication with D gives 37DD where A, B, C and D are different non-zero digits. What is the value of A+B+C?
(a) 18
(b) 16
(c) 15
(d) Cannot be determined due to insufficient data
10. If ABC and DEF are both 3-digit numbers such that A, B, C, D, E and F distinct and F are non-zero digits such that $ABC + DEF = 1111$, then what is the value of A+B+C+D+E+F?
(a) 28
(b) 29
(c) 30
(d) 31
11. Consider the following addition problem : $3P+4P+PP+PP=PQ2$; where P, Q and R are different digits.
What is the arithmetic mean of all such possible sums?
(a) 102
(b) 120
(c) 202
(d) 220
12. Consider the following multiplication problem:
 $(PQ) \times 3 = RQQ$, where P, Q and R are different digits and $R \neq 0$.
What is the value of $(P+R) \div Q$?
(a) 1
(b) 2
(c) 5
(d) Cannot be determined due to insufficient data
13. Three variables x, y and z take values 2, 3, 4 or 5 such that their values are always distinct. If M and N denote the largest possible value and the smallest possible value, respectively, for the expression $\{(x \times y) + z\}$; then M - N is
a) 11
b) 12
c) 13
d) 14
14. The 5-digit number PQRST (all distinct digits) is such that $T \neq 0$. P is thrice T. S is greater than Q by 4, while Q is greater than R by 3. How many such 5-digit numbers are possible?
(a) 3
(b) 4
(c) 5
(d) 6

15. In some code, letters P, Q, R, S, T represent numbers 4, 5, 10, 12, 15. It is not known which letter represents which number. If $Q-S = 2S$ and $T=R+S+3$, then what is the value of $P+R-T$?
- 1
 - 2
 - 3
 - Cannot be determined due to insufficient data
16. If the sum of the two-digit numbers AB and CD is the three-digit number 1CE, where the letters A, B, C, D, E denote distinct digits, then what is the value of A?
- 9
 - 8
 - 7
 - Cannot be determined due to insufficient data
17. A question is given followed by two Statements I and II. Consider the Question and the Statements and mark the correct option [UPSC 2025]
 Question: Let P, Q, R, S be distinct non-zero digits. If $PP \times PQ = RRSS$, where $P \leq 3$ and $Q \leq 4$, then what is Q equal to?
 Statement I: $R = 1$.
 Statement II: $S = 2$.
 Which one of the following is correct in respect of the above Question and the Statements?
- The Question can be answered by using one of the Statements alone, but cannot be answered using the other statement alone.
 - The Question can be answered by using either Statement alone.
 - The Question can be answered by using both the Statements together, but cannot be answered using either Statement alone.
 - The Question can be answered even without using any of the Statements.
18. A question is given followed by two Statements I and II. Consider the Question and the Statements and mark the correct option. [UPSC 2025]
 Question: Is $(p + q)^2 - 4pq$, where p, q are natural numbers, positive?
 Statement I: $p < q$.
 Statement II: $p > q$.
 Which one of the following is correct in respect of the above Question and the Statements?
- The Question can be answered by using one of the Statements alone, but cannot be answered using the other statement alone.
 - The Question can be answered by using either Statement alone.
 - The Question can be answered by using both the Statements together, but cannot be answered using either Statement alone.
 - The Question can be answered even without using any of the Statements.
19. Factorise 24
20. Factorise 144
21. Factorise 720
22. Find the number of factors of 24
23. Find the number of factors of 144